

Remote Sensing Thermometer

In construction project 'Remote Sensing Thermometer' published in June issue, the LM7905 regulator is shown with a 0.1 μ F capacitor across its output. While generally a 0.1 μ F capacitor is sufficient for stability across positive voltage regulator (7805) outputs, it is insufficient for negative voltage regulators. This can be checked in the datasheet of LM7905 from National Semiconductor and Fairchild Semiconductor. You need a minimum of 1 μ F tantalum or 25 μ F electrolytic across the output.

The use of a 0.1 μ F capacitor across the output of a negative regulator usually causes oscillations of several MHz, which are extremely undesirable and can be viewed with an oscilloscope. If oscillations are not occurring, it is just a matter of time and circumstances before they will occur.

Dr Ramkumar Ramaswamy
Through e-mail

AC-Powered LED Lamps

Regarding 'AC-Powered LED Lamps Without Rectifiers' circuit published in May issue, I wish to point out that:

1. It is ridiculous to provide isolation via a transformer but produce the dimming feature via a rotary switch on the mains side of the transformer. The same switch could be placed on the secondary side with smaller wattage resistors to produce a much more power-saver circuit.

2. The purpose of the two 100-ohm resistors, 100nF capacitor and 150mA fuse is unknown. These items are not needed.

3. The power lost in R2 is less than 5 mW, then why use a 2W resistor?

4. The 10-ohm resistors are far too small to have any effect.

The whole circuit is poorly designed and gives wrong information to the beginners.

Colin Mitchell
Australia

The author Petre Tzv. Petrov replies:
Thanks to the reader for his attention, opinion and queries. Answers to his

'Spot An Error' Award Winners

In 'Remote Sensing Thermometer' construction project published in June issue, pins 2 and 3 of IC7 have got interchanged.

Yadav Shivtej Vitthal

□ In the above-mentioned project, the value of C4 should be 1 μ F tantalum or 25 μ F electrolytic across the output.

Dr Ramkumar Ramaswamy

□ In Parts List of the 'Simple Protection For Audio Amplifier Loudspeakers' circuit idea published in June issue, the values of POT1 and POT2 are printed as 2.7 kilo-ohms instead of 4.7 kilo-ohms.

Sujit Koley

□ In the 'User-Friendly and Purposeful' innovation article published in June issue, under 'Listen to music the traditional way' the frequency response is misprinted as 10 kHz to 20 kHz instead of 10 Hz to 20 kHz.

Surendran V.K.

points are given below:

1. You can put the switch in the secondary side but that will require switch for a higher current. Also, there will be more difficulties in calculating the values of resistors in series with LEDs (lower values close to the resistance of the switch, internal resistance of the transformer, resistance of the LEDs, etc).

There are a lot of low-cost switches designed for the mains power supply. In the simplest case, you can use a simple on/off switch with two positions and a single resistor to reduce the input voltage.

2. Different people have different safety precautions and design approaches. More safety is always good from my point of view and the circuit ensures that. Smaller resistors such as an SMD may be used but that will make the implementation difficult and dangerous.

3. These components (R1 = 100E, R2 = 100E, C1 = 100nF) are useful especially if you live in an environment with a lot of electromagnetic noise in the mains. If you analyse the transitional processes in the circuit, you will find that they are useful. These components protect the transformer. If you do not need them, you can omit them.

4. The power loss is more than 5 mW. Also, some smaller resistors cannot work well on high voltages, especially over-voltage situations in the mains. Larger resistors are safer. You can put any resistor at your risk.

Resistors R6, R7 and R8 have some

effects. You can change their values if you are not satisfied. But the values depend on the transformer, LEDs and the maximum current through LEDs, forward resistance of LEDs, etc and can be from 10 ohms to more than 100 ohms, depending on the case.

The circuit is working reliably and safely. If you do not like it, you can send better solutions. There are so many ways to make LED lamps without rectifiers and this is only one of them.

Diodes' and Transistors' Market

The EFY Report on diodes' and transistors' market in June issue says, "Unfortunately, India is not a large producer of copper. This is due to the non-availability of copper ore in India." This statement is incorrect as copper ore is available in abundance in the mining areas of Rakha, Jharkhand, Malanjkhand in Madhya Pradesh, and Khetri and Kolihari in Rajasthan. See http://en.wikipedia.org/wiki/Category:Copper_mines_in_India

I am told that the reason for the low usage of domestic copper in electronics is its higher cost compared to the imported copper. Moreover, purity of the imported copper is tad higher than what is achieved locally.

Kartick Siddhartha
Hyderabad

EFY: The article was largely based on ELCOMOS report published by ELCINA some time ago. Thanks for pointing out the inaccuracies!